

TrustNode Scalability & Commercialization Architecture

Low-cost, secure, multi-customer, near-real-time rollout plan

PREPARED ON

22 April 2026

PREPARED FOR

TrustNode Product & Engineering

SCOPE

Minimal-change architecture evolution

1. Executive Summary

TrustNode already has the correct backbone for production: edge-first durable collection, outbox sync, ingest APIs, tenant-aware cloud schema, and live/historical split. The best strategy is to avoid a big rewrite and add a lean control plane for customer, edge, licensing, and URL tenancy management.

Strength

Edge-First Reliability

Local persistence and outbox allow operations during cloud/network outages.

Strength

Ingest Idempotency

Cloud ingest with scoped tokens and deduplication supports safe retries.

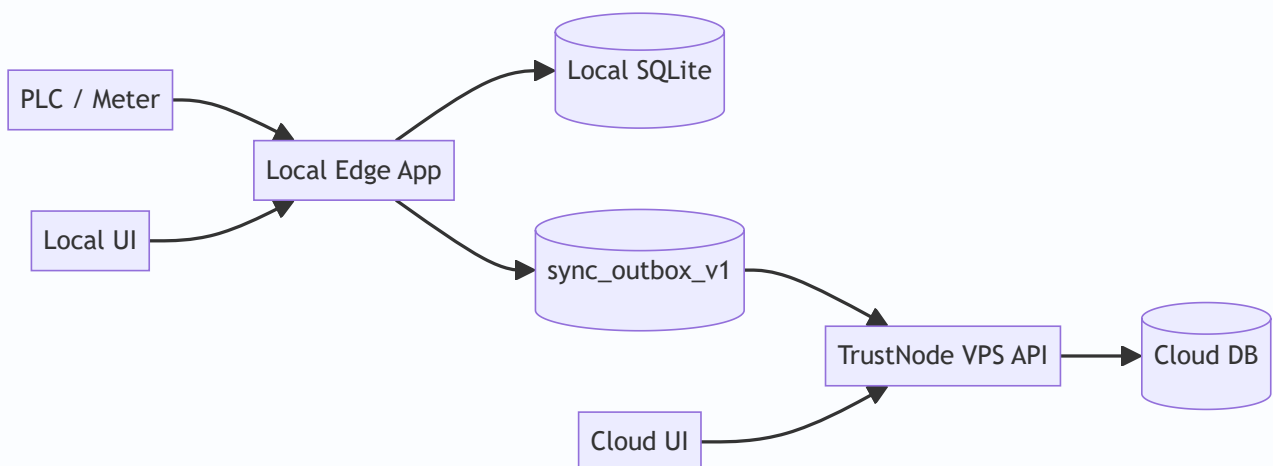
Gap

Commercial Control Plane

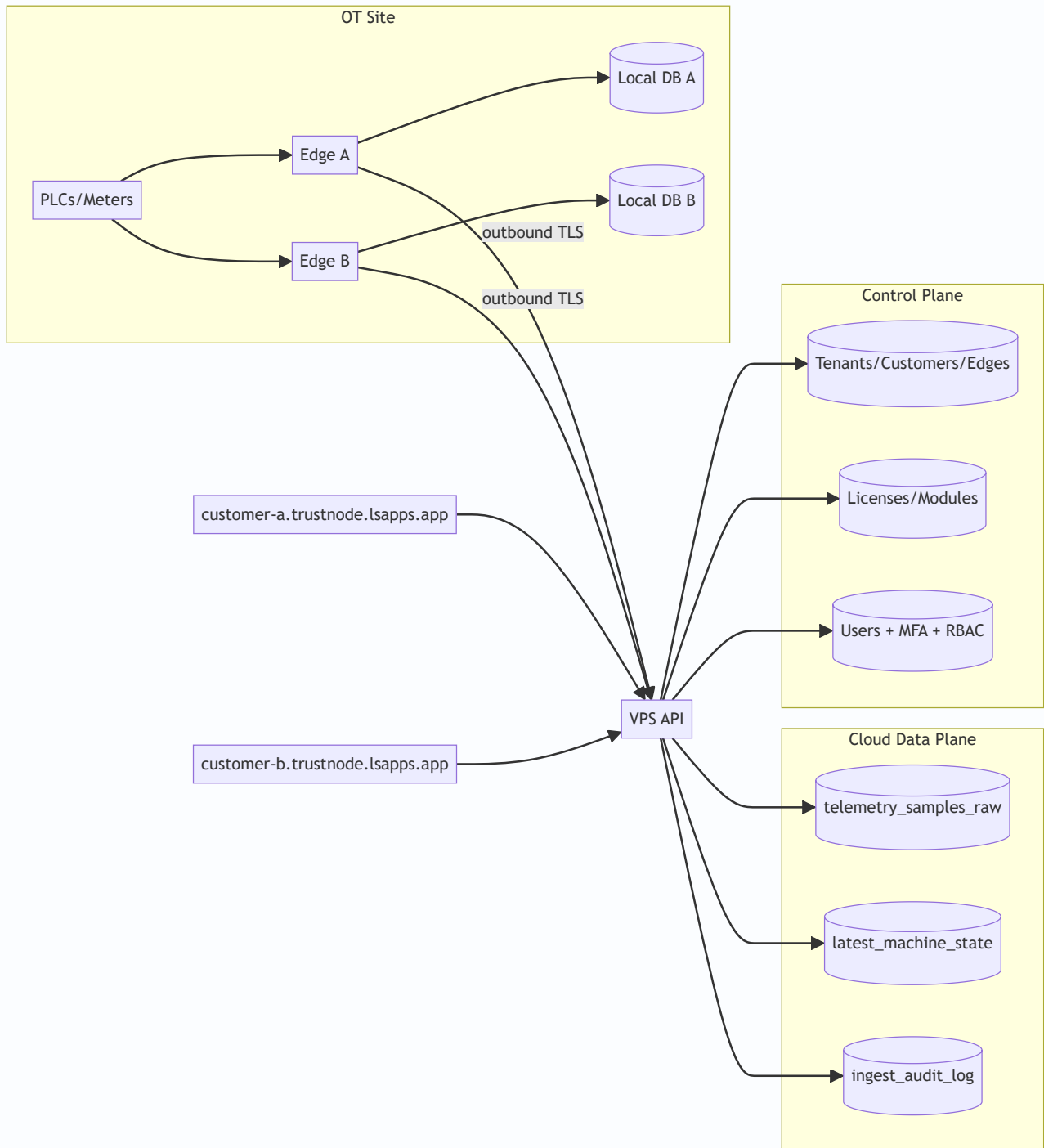
Need central tenant/customer/edge/license lifecycle and activation workflows.

2. Current vs Target Architecture

Current Simplified Data Flow

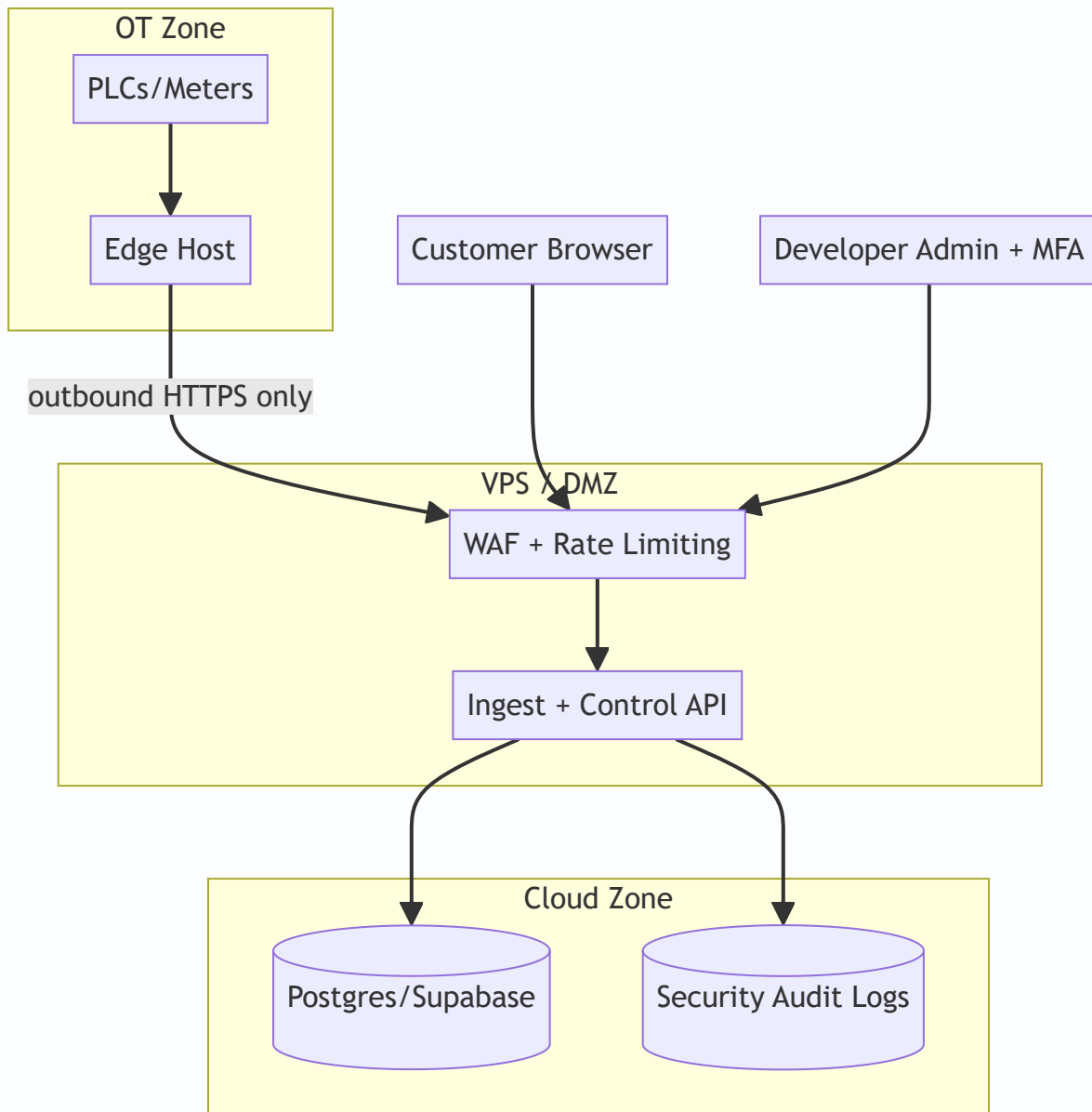


Target Three-Plane Model (Minimal Change)



3. Security & Isolation Topology

IT/OT Security Boundary (Zones and Conduits)



Mandatory Security Controls

- No inbound internet traffic into edge nodes.
- Separate human auth and machine/device auth.
- Tenant scope checks at token + API + query layers.
- RLS for all user-facing data tables.
- MFA for admin and privileged roles.

Data Protection Model

- Immutable raw telemetry for auditability.
- Latest snapshot for low-latency dashboards.
- Per-tenant logical separation and policy boundaries.
- PITR backups + restore drills for resiliency.

4. Customer URL and Licensing Strategy

Recommended: wildcard subdomain tenancy on your VPS/frontend proxy:

customer-a.trustnode.1sapps.app, customer-b.trustnode.1sapps.app

Map host header → tenant context in backend; deny cross-tenant access by default.

Control Plane Components

- Customers/Tenants registry
- Edges registry + activation workflow
- Licenses + module entitlements
- User memberships + role model
- Password recovery + security events

5. Rollout Plan (Minimal Risk)

Phase	Timeline	Goal	Key Deliverables
Phase 0	1-2 weeks	Stabilize current pipeline	Ingest URL/token diagnostics, startup checks, sync alarms
Phase 1	2-3 weeks	Control plane core	Tenant/customer/edge/license tables + APIs
Phase 2	2-3 weeks	Tenant URL routing	Wildcard subdomains, strict host-to-tenant binding
Phase 3	2-4 weeks	Security hardening	MFA enforcement, audit expansion, restore drills

6. Commercial Readiness Outcomes

Performance Targets

- Local edge live update: $\leq 1s$
- Cloud live update p95: $\leq 2s$
- Outbox depth bounded under normal connectivity
- No cross-tenant data exposure

Commercial Targets

- Per-customer onboarding and activation
- Per-module licensing enforcement
- Per-customer URL and role-based view control
- Admin-level recovery and account operations

7. Technical References

- PostgreSQL RLS: <https://www.postgresql.org/docs/current/ddl-rowsecurity.html>
- PostgreSQL partitioning: <https://www.postgresql.org/docs/current/ddl-partitioning.html>
- Supabase RLS guide: <https://supabase.com/docs/guides/database/postgres/row-level-security>
- Supabase API security: <https://supabase.com/docs/guides/api/securing-your-api>
- Supabase read replicas: <https://supabase.com/docs/guides/platform/read-replicas>
- Supabase custom domains: <https://supabase.com/docs/guides/platform/custom-domains>
- Supabase backups/PITR: <https://supabase.com/docs/guides/platform/backups>
- Supabase MFA: <https://supabase.com/docs/guides/auth/auth-mfa>

- NIST SP 800-82r3: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>
- OWASP API Security Top 10: <https://owasp.org/API-Security/editions/2023/en/0x11-t10/>
- Cloudflare wildcard DNS: <https://developers.cloudflare.com/dns/manage-dns-records/reference/wildcard-dns-records/>
- NGINX wildcard host routing: https://nginx.org/en/docs/http/server_names.html
- Let's Encrypt challenge types: <https://letsencrypt.org/docs/challenge-types/>

Generated from TrustNode architecture analysis - 22 April 2026.